

LLM TRAINING & SERVING

- LLM Fine-tuning
- LoRA / QLoRA
- Distributed Training
- LLM Evaluation
- OpenAI-Compatible Serving
- Multi-Task Adapters

MODELLING & ANALYTICAL METHODS

- Classical ML
- Deep Learning
- Computer Vision
- Natural Language Processing (NLP)
- Time-Series Forecasting
- Feature Engineering
- Behavioral Anomaly Detection
- Autoencoders
- Operations Research
- Vehicle Routing
- Mixed-Integer Programming
- Constraint Modelling
- Supply-Chain Optimization

FRAMEWORKS & APIS

- scikit-learn
- XGBoost
- PyTorch
- TensorFlow / Keras
- TensorFlow Serving
- Hugging Face Transformers / PEFT
- Unsloth
- vLLM
- Google OR-Tools

GPU & DISTRIBUTED COMPUTING

- CUDA GPU Libraries
- GPU-Accelerated Computing
- RAPIDS cuDF / CuPy

EXPERIMENTATION & MODEL LIFECYCLE

- MLflow
- Experiment Tracking
- Model Evaluation
- Model Versioning / Registry
- Automated Retraining
- Vertex AI Pipelines
- Azure ML Pipelines
- Vertex AI Feature Store

CLOUD & SERVERLESS SERVICES

- GCP
- Microsoft Azure
- GKE
- Vertex AI
- Azure ML
- Cloud Run
- Cloud Functions
- Azure Function Apps
- Logic Apps

DEPLOYMENT & PRODUCTION RELIABILITY

- ML System Architecture
- Cloud-Native ML Systems
- Docker
- Kubernetes
- Helm
- Seldon Core
- Istio
- HPA Autoscaling
- CI/CD
- Cloud Build
- Azure DevOps / Pipelines
- ArgoCD
- Google KCC
- GitOps
- JWT Authentication

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

WORK EXPERIENCE

- Lead Artificial Intelligence Engineer** Best Buy | Jul 2025 – Present
 - Engineered a configuration-driven **LLM fine-tuning framework** supporting distributed training, task-specific adapters, and automated evaluation. Standardized task configuration and model artifacts enabled multiple tasks to share a common serving deployment.
 - Deployed **self-hosted LLM serving** with vLLM on GKE and **estimated annual API-cost savings of ~\$300K**, keeping model inputs within company-managed infrastructure and reducing exposure to external API availability and behavior during peak demand.
 - Built **production fraud-detection systems** spanning multiple models and batch, streaming, and real-time components, including a feature pipeline handling **40,000 events/sec** at peak. Supported a **200 ms scoring SLA** and improved fraud-score coverage by **~20%** during holiday traffic.
 - Lead architecture and production delivery of reusable LLM infrastructure; partner with data scientists to deploy secure, monitored retail AI services.
- Lead ML Engineer — ML Engineering & Operations** Wipro | Aug 2023 – Jul 2025
 - Engineered **AWS account digital fingerprinting** on NVIDIA DGX infrastructure using GPU-accelerated data processing. Trained per-entity behavioral autoencoders on **billion-row security datasets** to detect anomalies, with continuous retraining and model management.
 - Scaled **Proofpoint phishing-detection pipelines** to **2,400 emails/sec**, double the previous rate. Incorporated analyst feedback and Microsoft Sentinel integration to identify phishing missed by initial filters and support quarantine response.
 - Automated **abuse-inbox triage** with ML routing and OCR extraction across email bodies and attachments. Routed scammer identifiers to security teams and mail partners for takedowns, **doubling the prior case volume** and reducing manual review.
 - Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery; screened **300+ resumes** and conducted **50+ technical interviews**. Received Best Buy AML Hall of Fame recognition (2024) during the Wipro client engagement.

MONITORING, TESTING & REPORTING

Prometheus Grafana
Google Cloud Monitoring
Model Monitoring Load Testing
Unit / Integration Testing Streamlit
Power BI

LANGUAGES, DATA & STORAGE

Python SQL Apache Beam / Dataflow
Pub/Sub BigQuery MS SQL Server
Redis Caching GCS
Azure Blob Storage Artifact Registry
Pandas / NumPy Dask Git
Linux / Shell Scripting

CERTIFICATIONS

Google Cloud Certified
Generative AI Leader

Oracle Cloud Infrastructure 2024
Generative AI Certified Professional

Microsoft Certified
Azure AI Engineer Associate

Microsoft Certified
Azure Data Scientist Associate

Microsoft Certified
Power BI Data Analyst Associate

EDUCATION

B.E. Electrical & Electronics
Andhra University, Visakhapatnam
2012 – 2016 · 80%

HONORS & AWARDS

Best Buy — Talent Quotient 2026
Fraud-score coverage +20% · ~\$300K
estimated annual API-cost savings

Best Buy client - AML Hall of Fame
2024 (Wipro engagement)
Above & Beyond recognition

Wipro — 6 further awards
Above & Beyond · Pinnacle · Being
Responsive x3 (2024–25)

Accenture
Delivery Excellence · Kudos x2

Infosys
Insta Award x2

DOMAIN & CLIENT BREADTH

Retail & e-commerce · Cybersecurity ·
Supply chain & logistics · Semiconductor
manufacturing · Market intelligence (15+
verticals) · Insurance.

Clients across the US, Norway, South
Korea, and the UK.

LEADERSHIP & DELIVERY

ML system architecture · Team
leadership & mentoring · Technical hiring
· Cross-functional delivery · Stakeholder
management.

- Senior Data Scientist — Data Science / Azure ML

Accenture | May 2021 – Sep 2023

 - Developed **product-quality prediction for BamaGruppen** on Azure ML, with automated retraining and stakeholder alerts. Increased truck inspections from **130 to 160/day**, defect detection by **17%**, and customer approval by **20%**, delivering **\$2.5M annual savings**.
 - Designed **last-mile vehicle routing** with OR-Tools under capacity, delivery-window, waiting-time, and store-demand constraints. Compared optimized routes with manual plans to improve resource utilization and reduce logistics costs.
 - Implemented **transport optimization** using mixed-integer assignment models and APIs, automating daily dispatch and replacing manual Excel planning.
 - Modelled **supply-chain and distribution-hub scenarios** across cost, freshness, carbon footprint, and warehouse volumes to support network-design decisions.
 - Led cross-functional delivery, mentored data scientists, and presented recommendations to Norwegian stakeholders.
- Data Scientist

GD Research Centre Pvt Ltd | May 2020 – May 2021

 - Developed domain-aware **GlobalData forecasts** and scenario analysis across **15+ industry verticals**, informing market-intelligence reports and resource-allocation decisions.
- Deep Learning Engineer

Tata Consultancy Services | Jul 2019 – Mar 2020

 - Built and deployed an **end-to-end ML system for semiconductor defect detection**, using distributed training on **~1 million images in ~3 hours** and achieving **90% classification accuracy**. Deployed the system at Amkor in South Korea and trained the client team to operate it.
- Senior Software Engineer — ML

Infosys | May 2016 – Jul 2019

 - Implemented **opportunity win/lose prediction** with logistic-regression scoring and data pipelines, integrating predictions into business dashboards for sales prioritization.
 - Delivered tested **insurance ID-card generation** workflows, from impact analysis and design through implementation, support, and team knowledge sharing.

OPEN-SOURCE PROJECT

vLLM Model Chooser: Built an interactive GPU-to-model sizing tool with VRAM, quantization, and KV-cache fit filters. [Live project.](#)